Exercise Sheet 23

The notation is that of Lecture 23. We use the isotropic basis $e_{\pm j}$ and e_0, the orthonormal basis u_a, the contraction c, the element $q = \sum_a u_a^2$, and the Hodge operator $\star$ with the volume form of the lecture. Exercises 23.1–23.3 are core. Exercises 23.4–23.6 practise transfer.

Exercise 23.1 (The vector square of $\mathfrak{so}_5$). Let $N = 5$, so $n = 2$ and the type is B_2.

(i) Express q in the basis $e_{\pm 1}, e_{\pm 2}, e_0$.

(ii) Prove $5 \otimes 5 = 1 \oplus 10 \oplus 14$. Give the highest weights in Dynkin labels.

(iii) Derive $\dim L(a\omega_1 + b\omega_2) = (a+1)(b+1)(a+b+2)(2a+b+3)/6$ and check 14 and 10.

(iv) Check the zero-weight multiplicities of both sides.

Exercise 23.2 (A middle exterior power).

(i) Let $N = 4$, so $n = 2$. This lies outside the lecture's standing hypothesis $n \geq 3$. The two eigenvector computations in the Hodge eigenvalue proof use only linear algebra and diagonal weight spaces, so they apply here. The identification with highest-weight summands is not needed. Compute $\star(e_1 \wedge e_2)$ and $\star(e_1 \wedge e_{-2})$ directly in the orthonormal basis. Compare with c_2.

(ii) Let $N = 6$. Decompose $\Lambda^3 V$ and find the eigenvalue of $\star$ on each summand. Show that $J = \mathrm{i}\star$ is an involution.

(iii) What happens to the eigenspaces of J if vol is replaced by $-$vol?

Exercise 23.3 (A second proof of the harmonic decomposition). Let $f \in \operatorname{Sym}^k V$.

(i) Prove $c(qf) = q\,c(f) + (N + 2k)f$.

(ii) For $h \in \operatorname{Sym}^k_0 V$ and $j \geq 1$, prove $c(q^j h) = j(N + 2k + 2j - 2)\, q^{j-1} h$.

(iii) Without the Hermitian form, prove $\operatorname{Sym}^k V = \bigoplus_{0 \leq j \leq k/2} q^j \operatorname{Sym}^{k-2j}_0 V$.

(iv) Find the trace-free part of $e_1 e_{-1}$.

Exercise 23.4 (Dimension checks).

(i) Verify $\dim \operatorname{Sym}^3 \mathbb{C}^5 = \dim L(3\omega_1) + \dim L(\omega_1)$ for $\mathfrak{so}_5$.

(ii) Verify $\dim \operatorname{Sym}^4 \mathbb{C}^8 = \dim L(4\omega_1) + \dim L(2\omega_1) + 1$ for $\mathfrak{so}_8$.

(iii) Identify $\Lambda^3 \mathbb{C}^8$ and $\Lambda^5 \mathbb{C}^8$ for $\mathfrak{so}_8$.

Exercise 23.5 (Missing and present weights).

(i) For $\mathfrak{so}_7$, decide which of $L(\omega_3)$, $L(2\omega_3)$, $L(\omega_1 + \omega_3)$ and $L(\omega_1 + 2\omega_3)$ are submodules of some $V^{\otimes k}$. Give the least such k.

(ii) In D_n, prove that every weight μ of $V^{\otimes k}$ satisfies $\sum_j \mu_j \equiv k \pmod 2$. Deduce that, for $\mathfrak{so}_8$, $V^{\otimes 2}$ has no submodule isomorphic to 8_v, and $V^{\otimes 3}$ has no submodule isomorphic to $L(2\omega_1)$.

(iii) Show that the parity rule of (ii) fails in B_n.

Exercise 23.6 ($\Lambda^2 V$ is the adjoint module). For $u, v \in V$ define the operator $M_{u,v} \in \mathfrak{gl}(V)$ by

$$M_{u,v}(w) = (v, w)u - (u, w)v \qquad (w \in V).$$

(i) Prove $M_{u,v} \in \mathfrak{so}(V)$.
(ii) Prove $[X, M_{u,v}] = M_{Xu,v} + M_{u,Xv}$ for $X \in \mathfrak{so}(V)$.
(iii) Prove that $u \wedge v \mapsto M_{u,v}$ is an isomorphism $\Lambda^2 V \to \mathfrak{so}(V)$ of modules.
(iv) Find the image of $e_1 \wedge e_2$.